EchoProof

Crowdsourced, verifiable environmental sensing rewarded on Solana



The Problem & Our Solution

Environmental and urban data is costly to collect, often centralized, stale, or unverifiable, leaving cities and researchers without reliable real-time insight.

- Noise, air quality, and GPS data are expensive to gather at city scale
- Existing datasets are centralized, outdated, or easy to fake
- Smartphones are everywhere but their sensor data is untapped
- EchoProof turns phones into a verifiable DePIN sensing network on Solana
- Contributors earn tokens; buyers get trustworthy, queryable data

How It Works

A simple flow: capture, prove, reward, and query, all anchored by Solana smart contracts and decentralized storage.

- User flow: app passively captures noise, GPS, or light data in background
- Data is signed with device attestation to prevent spoofing
- Proofs and metadata are submitted on-chain; raw data stored on IPFS/Arweave
- Solana smart contract verifies proofs and distributes token rewards
- Buyers query the marketplace for verified datasets by region and time

Built with

Solana, React Native, Anchor, Rust, IPFS/Arweave, Solana Pay, PostgreSQL for indexing

Why Now & What's Next

DePIN is booming on Solana, and cheap sensors plus low fees finally make crowdsourced verified data practical.

- Solana's low-cost transactions make per-contribution rewards economically viable
- DePIN is one of the fastest-growing categories in the ecosystem right now
- Next: pilot data purchase with a city or university partner
- Next: add hardware sensor support like air quality dongles
- Next: reputation-weighted rewards to improve long-term data quality

Category

DePIN